MILITARY SPECIFICATION

CARTRIDGE, IGNITION, XM299
PARTS FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers parts for XM299, Ignition Cartridges for 81mm, HE, Mortar Ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS AND STANDARDS

MILITARY

MIL-STD-1169	-Packaging, packing and marking for Shipment of Inert Ammunition Components.
MIL-A-48078	-Ammunition, Standard Quality Assurance Provisions, General Specification for

DRAWINGS

PRODUCT AND PACKAGE DRAWINGS

9293426	-Plug Firing
9285355	-Tube, Paper
9293425	-Liner
9280537	-Cap
9293424	-Tube, Flash
9293428	-Body
9293427	-Head
9243255	-Paper, Cartridge

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3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on Drawings (dwgs) 9293426, 9285355, 9293425, 9293424, 9293428, 9293427, 9243255, and 9280537, as applicable, and with all requirements specified in applicalbe specifications.

3.3 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Bursting strength.-The tube assemblies shall not burst at a compression load of 57 pounds, minimum , but shall burst at a compression load of 192 pounds, maximum. Testing shall be as specified in 4.4.3.1.

3.5 Workmanship.-All parts shall be fabricated and finished in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspection.-The following types of inspection shall be conducted on this item:

a. First Article Inspection.
b. Quality Conformance Inspection.

4.3 First Article Inspection

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4.3.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of twenty-five (25) complete sets of parts.

4.3.2 Inspections to be performed.-The First Article parts submitted in accordance with paragraph 4.3.1 will be subjected by the Government to any or all the requirements of the applicable drawings and to the inspections and tests listed in paragraphs 4.4.2 through 4.4.3.

4.3.3 Rejection.-See MIL-A-48078(PA)

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with lot formation provisions of MIL-A-48078.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling Plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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Quality Conformance Inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9285355
4.4.2.1	Tube, Paper				NEXT HIGHER ASSEMBLY 9285390
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A 101	Thickness		0.40%	3.2	Gage Note 1
Major B 131 132	Length Inside Diameter		0.40% 0.40%	3.2 3.2	Gage Gage
Minor	None defined				

NOTES:

1. Inspection equipment design approval delegated to DCAS.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Liner		9293425
			NEXT HIGHER ASSEMBLY 9293422

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A 101	Thickness		0.40%	3.2	Gage Note 1
Major B 131 132	Outside diameter Length		0.40% 0.40%	3.2 3.2	Gage Gage
Minor	None defined				

NOTES:

1. Inspection equipment design approval delegated to DCAS.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48155(PA)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9293428
4.4.2.3	Body				NEXT HIGHER ASSEMBLY 9293422
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major A					
101.	Flash hole missing or obstructed		100%	3.2	Visual
Major B					
131.	Pitch diameter of external threads, min		0.40%	3.2	Gage
132.	Major diameter of external threads, min		0.40%	3.2	Gage
133.	Diameter of primer cavity		0.40%	3.2	Gage
134.	Depth of primer cavity		0.40%	3.2	Gage
135.	Depth of flash tube cavity		0.40%	3.2	Gage
136.	Diameter of flash tube cavity		0.40%	3.2	Gage
137.	Diameter of flash hole		0.40%	3.2	Gage
138.	Large diameter		0.40%	3.2	Gage
139.	Diameter of pellet cavity		0.40%	3.2	Gage
140.	Depth of pellet cavity		0.40%	3.2	Gage
141.	Length overall		0.40%	3.2	Gage
142.	True position of primer cavity		0.40%	3.2	Gage
143.	Width of crimp groove		0.40%	3.2	Gage
144.	Depth of crimp groove		0.40%	3.2	Gage
145.	Small diameter		0.40%	3.2	Gage
146.	Diameter of countersink at base of primer cavity, max		0.40%	3.2	Gage
147.	Length from large diameter to small end		0.40%	3.2	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48155(PA)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.3	Body		9293438
			NEXT HIGHER ASSEMBLY 9293422

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major B					
148.	Length from large diameter to end of external thread		0.40%	3.2	Gage
149.	Length of external thread, max		0.40%	3.2	Gage
150.	Diameter of thread undercut		0.40%	3.2	Gage
151.	Length of large diameter section		0.40%	3.2	Gage
152.	Length from large diameter end to crimp groove		0.40%	3.2	Gage
153.	Evidence of protective finish		0.40%	3.2	Visual
Minor					
201.	Diameter of countersink at outer end of primer cavity, max		0.65%	3.2	Gage
202.	Pitch diameter of external thread, max		0.65%	3.2	Gage
203.	Major diameter of external thread, max		0.65%	3.2	Gage
204.	Radii, chamfer or countersink missing		0.65%	3.2	Visual
205.	Surface finish improper		0.65%	3.2	Visual
206.	Evidence of poor workmanship(see 3.5)		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48155(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9293426
4.4.2.4	Plug, Firing				NEXT HIGHER ASSEMBLY 9293422
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major A					
101.	Length of tip		0.40%	3.2	Gage Note 1
102.	Total length including tip		0.40%	3.2	Gage Note 1
103.	Length from tip to small diameter		0.40%	3.2	Gage Note 1
104.	Radius improper		0.40%	3.2	Gage Note 1
Major B					
131.	Large diameter		0.40%	3.2	Gage
132.	Small diameter		0.40%	3.2	Gage
133.	Evidence of protective finish		0.40%	3.2	Visual
Minor					
201.	Surface finish improper		0.65%	3.2	Visual
202.	Evidence of poor workmanship (see 3.5)		0.65%	3.2	Visual

NOTES:

1. Inspection equipment design approval delegated to DCAS.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48155(PA)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.5	Head		9293427
			NEXT HIGHER ASSEMBLY 9293422

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
131.	Pitch diameter of internal threads, max		0.40%	3.2	Gage
132.	Minor diameter of internal threads, max		0.40%	3.2	Gage
133.	Pitch diameter of external threads, min		0.40%	3.2	Gage
134.	Major diameter of external threads, min		0.40%	3.2	Gage
135.	Length of internal thread, min		0.40%	3.2	Gage
136.	Overall length		0.40%	3.2	Gage
137.	Length of large internal diameter		0.40%	3.2	Gage
138.	Diameter of firing plug hole		0.40%	3.2	Gage
139.	Large diameter		0.40%	3.2	Gage
140.	Small diameter		0.40%	3.2	Gage
141.	Diameter of thread undercut (two places)		0.40%	3.2	Gage
142.	Length of thread undercut (two places)		0.40%	3.2	Gage
143.	Evidence of protective finish		0.40%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48155(PA)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9293427
4.4.2.5	Head				NEXT HIGHER ASSEMBLY 9293422
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201.	Pitch diameter of internal thread, min		0.65%	3.2	Gage
202.	Pitch diameter of external thread, max		0.65%	3.2	Gage
203.	Major diameter of external thread, max		0.65%	3.2	Gage
204.	Diameter of large end		0.65%	3.2	Gage
205.	Length of large diameter		0.65%	3.2	Gage
206.	Length from large diameter to external thread		0.65%	3.2	Gage
207.	Length from large diameter to beyond external thread		0.65%	3.2	Gage
208.	Radii, chamfers or countersinks missing		0.65%	3.2	Visual
209.	Surface finish improper		0.65%	3.2	Visual
210.	Diameter of spanner holes		0.65%	3.2	Gage
211.	True position of spanner holes		0.65%	3.2	Gage
212.	Evidence of poor workmanship (see 3.5)		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48155 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Tube, Flash		9293424
			NEXT HIGHER ASSEMBLY
			9293428

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major A					
101.	Longitudinal flash hole missing or obstructed		100%	3.2	Visual
102.	Flash hole(s) missing or excessive in number		100%	3.2	Visual
103.	True position of flash holes		0.40%	3.2	Gage
104.	Diameter of flash holes		0.40%	3.2	Gage
Major B					
131.	Length		0.40%	3.2	Gage
132.	Large outside diameter		0.40%	3.2	Gage
133.	Small outside diameter		0.40%	3.2	Gage
134.	Length of small outside diameter		0.40%	3.2	Gage
135.	Large inside diameter		0.40%	3.2	Gage
136.	Length of large inside diameter		0.40%	3.2	Gage
137.	Evidence of protective finish		0.40%	3.2	Visual
138.	Small inside diameter				Gage
Minor					
201.	Chamfers missing		0.65%	3.2	Visual
202.	Surface finish improper		0.65%	3.2	Visual
203.	Evidence of poor workmanship (see 3.5)		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48155(PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Cap		9280537
			NEXT HIGHER ASSEMBLY 9293422

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101.	Outside diameter		0.40%	3.2	Gage Note 1
Major B					
131.	Total length		0.40%	3.2	Gage
132.	Depth of groove		0.40%	3.2	Gage
133.	Width of groove		0.40%	3.2	Gage
134.	Diameter of cavity		0.40%	3.2	Gage
135.	Depth of cavity		0.40%	3.2	Gage
Minor					
201.	Radii or chamfers missing		0.65%	3.2	Visual
202.	Surface finish improper		0.65%	3.2	Visual
203.	Evidence of poor workmanship (see 3.5)		0.65%	3.2	Visual

NOTES:

1. Inspection equipment design approval delegated to DCAS.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48155(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Packed Carton prior to sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201.	Support(s) missing or damaged		0.65%	5.1	Visual
202.	Divider(s) missing or damaged		0.65%	5.1	Visual
203.	Improperly packed		0.65%	5.1	Visual
204.	Evidence of poor workmanship (see 3.5)		0.65%	5.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48155(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Carton after sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201.	Contents loose		0.65%	5.1	Manual
202.	Cartons damaged (cut or damaged through all layers)		0.65%	5.1	Visual
203.	Carton not properly sealed		0.65%	5.1	Visual
204.	Marking, missing, incorrect or illegible		0.65%	5.1	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Bursting strength (see 3.4) Major defect.-See Figure 1.

4.4.3.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan and procedure for this test shall be in accordance with MIL-STD-105 Code Letter J and AQL of 0.65 percent. The test shall be performed as specified in 4.5.1.

4.4.3.1.2 After three consecutive lots have met the criteria of 4.3.3.1.1, the sampling plan for this test shall be in accordance with MIL-STD-105 Code Letter F, using an AQL of 0.65 percent. The test shall be performed as specified in 4.5.1.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Bursting strength.-The tube and liner (without Milar tape) shall be cut in half lengthwise and one of the combined halves placed in the fixture (see figure 1). The stop (3) shall be completely removed or backed off so that it does not touch the punch (2). Place fixture in universal tester set for a travel speed of 0.125 inches per minute. Proceed with Test Number 1 and 2.

TEST NUMBER 1: NO BURST

Apply 57 pound compression load, remove body assembly. Item is rejected if evidence of bursting is observed. Tearing or puncturing of material shall be considered evidence of bursting.

TEST NUMBER 2: BURSTING

If item is not rejected, replace it in fixture, using opposite end of body and apply 192 pounds compression load. If item shows no evidence of bursting it shall be rejected.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.-Packaging, packing and marking shall be in accordance with the requirements of Level C in MIL-STD-1169.

6. NOTES

6.1 Intended Use.-The components covered by this specification are intended for use on Cartridge, Ignition, XM299.

6.2 Ordering Data.-See MIL-A-48078 (PA).

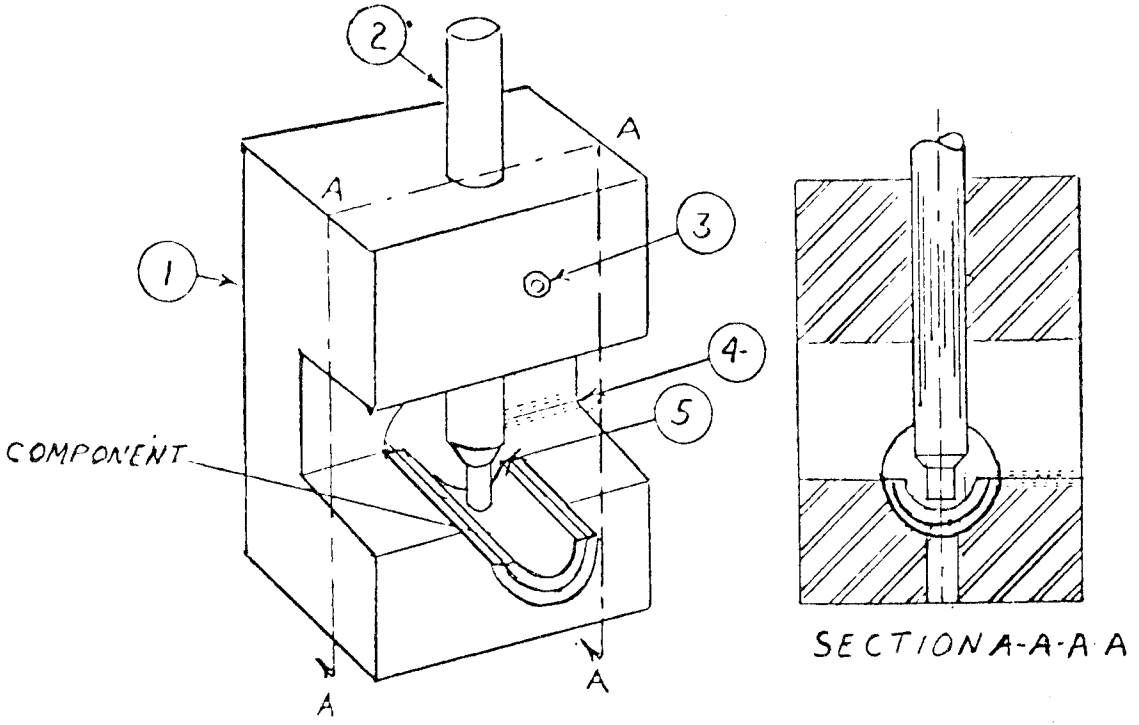
6.3 Submission of Designs for Approval.-See MIL-A-48078. Submit equipment designs, as required, to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

Custodian:
ARMY-PA

Preparing Activity:
ARMY-PA

Project Number: 1310-A-017

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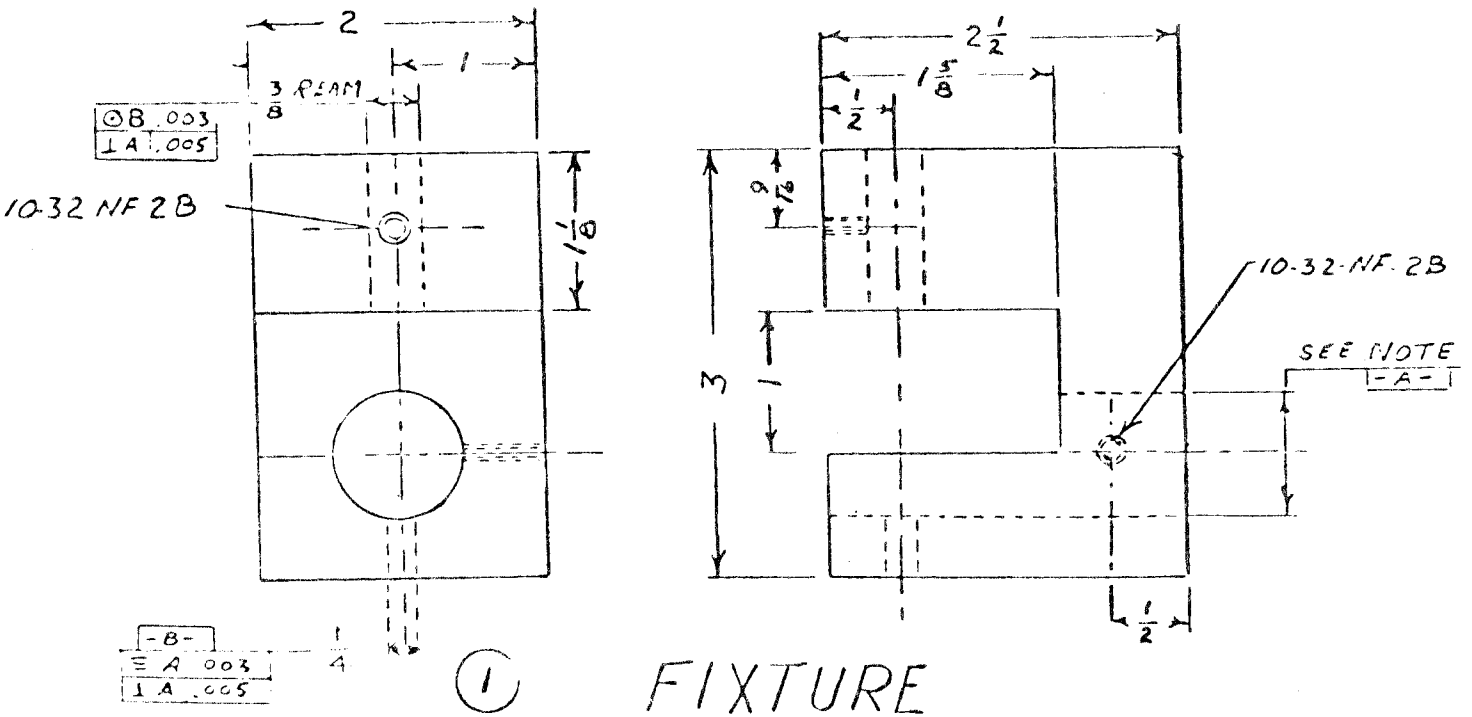


ASSEMBLY
BURSTING STRENGTH
TEST FIXTURE

FIGURE 1 SHEET 1

NIZ-C-48155 (PA)

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(1) FIXTURE

MAT'L - MACH. STEEL 1 REQ'D

(3) 10-32 SOC. SETSCREW FULL DOG PT. 3/8 LG.

(4) 10-32 NF-2A SOC. SETSCREW FULL DOG PT. 1/2 LG

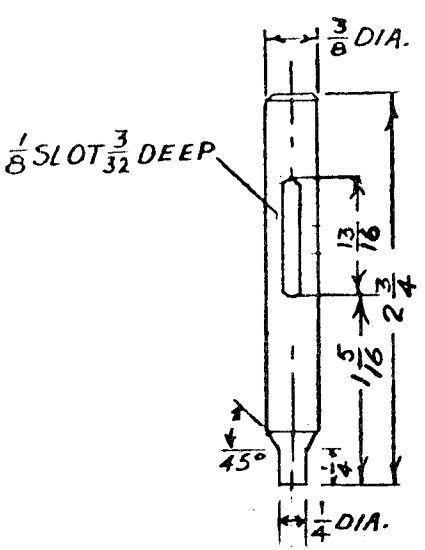
NOTE: DIAMETER TO SUIT APPLICABLE BODY ASS'Y

FIGURE 1 SHEET 2

DIMENSIONS ARE IN INCHES

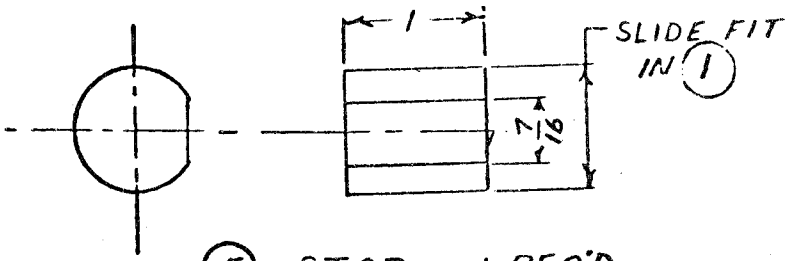
TOLERANCES ON - DECIMALS ± .005
FRACTIONS ± 1/64
ANGLES ± 2°

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(2) PUNCH

MATERIAL, DRILL ROD
1 REQ'D.



(5) STOP 1 REQ'D

MAT'L - MACH. STEEL

FIGURE 1 SHEET 3

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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